

Business Analytics Capstone

MIS 382

General Motors

**Muhammad Ibrahim, Sarah Lee, John Hwang, Rohan Giri, Varsha Manju
Jayakumar, Ammar Mustufa**

Assignment # 3

ROI Calculation for the Project

Costs:

Development Costs: \$250,000 (data acquisition, model development, and dashboard creation)

Installation Costs: \$4,000,000 (installation of 100 chargers at an average cost of \$40,000 per charger. We will use level 3 DC fast charger as per client requirements)

Annual Maintenance Costs: \$200,000 (Rental charges), \$150,000 (Maintenance)

Benefits:

Projected increase in EV sales: \$1,200,000 annually (from increased adoption rates and new market penetration. Upsell of vehicles). Annual growth at 30%

Charger utilization revenue: \$500,000 annually (charging fees collected from public use)
Annual growth at 30%

Environmental impact: Reduction of 2,000 tons of CO2 annually (estimated social benefit valued at \$200,000 per year).

Improved customer satisfaction: Increased brand loyalty and market share (qualitative benefit).

EV Charger Placement ROI Calculation

1. Traditional ROI (1-year):

$$\text{ROI} = (\$1,700,000 - \$350,000) / \$4,000,000 \times 100 = 27\%$$

2. Net Present Value (NPV, 3-year projection, 10% discount rate):

Assume the following data for the NPV calculation:

- Initial Investment (Development + Installation): \$4,250,000
- Annual Projected Benefit (Sales plus service): \$1,200,000 + \$500,000 (Annual Growth at 30%)
- Annual Maintenance Costs: \$350,000 (Annual Increase at 5%)

$$\begin{aligned} \text{NPV} = & (\$1,700,000 - \$350,000) / (1+0.1)^1 + \\ & (\$2,210,000 - \$367,500) / (1+0.1)^2 + \\ & (\$2,873,000 - \$385,875) / (1+0.1)^3 \end{aligned}$$

$$= \$461,8613.82 - \$4,250,000$$

$$\text{NPV} = \$368,613.82$$

3. Payback Period

Initial Investment / Annual Cash flow

Years:

$$(\$1,700,000 - \$350,000) + (\$2,210,000 - \$367,500) + (\$2,873,000 - \$385,875) = \$461,8613.82$$

3 years

Months:

$$(\$461,8613.82 - \$4,250,000) / \$4,250,000$$

= 0.87 years

3.87 years